

4.2.14

EE25BTECH11011 - Navya

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Question

Find the direction and normal vector for the line

$$x + y = 0 \quad (1)$$

Solution

The equation of line is :

$$x + y = 0 \quad (2)$$

This equation can be expressed in terms of matrices
Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (3)$$

$$\mathbf{n}^T = \begin{pmatrix} 1 & 1 \end{pmatrix} \quad (4)$$

$$c = 0 \quad (5)$$

The line equation can be written as:

$$\mathbf{n}^T \mathbf{x} = c \quad (6)$$

Solution

Where $\mathbf{n}$ is the normal vector of the given line

The direction vector of the line can be found by observing the normal vector.

$$\mathbf{m} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (7)$$

This is true because if the director vector is represented as

$$\mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (8)$$

then the normal vector can be represented as

$$\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix} \quad (9)$$

Solution

This can be verified by the following equation:

$$\mathbf{n}^T \mathbf{m} = 0 \quad (10)$$

$$\begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 0 \quad (11)$$

The normal vector of the line is $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

The director vector of the line is $\mathbf{m} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

From the figure, it is clearly verified that the theoretical solution matches with the computational solution.

Part 1

```
#include <stdio.h>

/* Dot product of 2D vectors */
int dot_product(int a[2], int b[2]) {
    return a[0]*b[0] + a[1]*b[1];
}

/* Check if two vectors are orthogonal */
int is_orthogonal(int a[2], int b[2]) {
    return dot_product(a, b) == 0;
}

/* Equation of the line:  $x + y = 0 \rightarrow y = -x$  */
double line_equation(double x) {
    return -x;
}
```

Part 2

```
/* Example test */
int main(void) {
    int n[2] = {1, 1}; // normal vector
    int m[2] = {-1, 1}; // direction vector

    printf("Dot product of n and m = %d\n", dot_product(n, m));
    printf("Are n and m orthogonal? %s\n", is_orthogonal(n, m) ?
        "Yes" : "No");

    for (double X = -5; X <= 5; X += 2.5) {
        printf("For x = %.2f, y = %.2f (on line x+y=0)\n", X,
            line_equation(X));
    }
    return 0;
}
```

Part 1

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load compiled shared library
lib = ctypes.CDLL("./code5.dll")

# Define signatures
lib.dot_product.argtypes = [ctypes.POINTER(ctypes.c_int), ctypes.POINTER(ctypes.c_int)]
lib.dot_product.restype = ctypes.c_int

lib.is_orthogonal.argtypes = [ctypes.POINTER(ctypes.c_int), ctypes.POINTER(ctypes.c_int)]
lib.is_orthogonal.restype = ctypes.c_int
```


Part 2

```
lib.line_equation.argtypes = [ctypes.c_double]
lib.line_equation.restype = ctypes.c_double

# --- Line:  $x + y = 0$  ---
normal_vector = (ctypes.c_int * 2)(1, 1) # n = (1, 1)
direction_vector = (ctypes.c_int * 2)(-1, 1) # m = (-1, 1)
vector_origin = np.array([0, 0])

# Dot product + orthogonality check
dp = lib.dot_product(normal_vector, direction_vector)
print(f"Dot product of n and m: {dp}")

if lib.is_orthogonal(normal_vector, direction_vector):
    print("The vectors are orthogonal (as expected).")
else:
    print("The vectors are NOT orthogonal.")
```

Part 3

```
# Compute points on line  $y = -x$ 
x_vals = np.linspace(-5, 5, 200)
y_vals = [lib.line_equation(float(x)) for x in x_vals]

# --- Plotting ---
plt.style.use('seaborn-v0_8-whitegrid')
plt.figure(figsize=(8, 8))

# Line
plt.plot(x_vals, y_vals, label='Line:  $x + y = 0$ ', color='blue',
         zorder=1)

# Direction vector
plt.quiver(vector_origin[0], vector_origin[1],
           direction_vector[0], direction_vector[1],
           angles='xy', scale_units='xy', scale=1,
           color='green', label='Direction Vector', zorder=2)
```

Part 4

```
# Normal vector
plt.quiver(vector_origin[0], vector_origin[1],
           normal_vector[0], normal_vector[1],
           angles='xy', scale_units='xy', scale=1,
           color='red', label='Normal Vector', zorder=2)

# Origin point
plt.plot(vector_origin[0], vector_origin[1], 'o', color='purple',
         markersize=8,
         label='Origin (0, 0)')

plt.title('Line  $x + y = 0$  with Direction and Normal Vectors')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Part 5

```
plt.axis('equal')
plt.legend()
plt.grid(True)

plt.xlim(-5, 5)
plt.ylim(-5, 5)

# Save figure
plt.savefig("fig5.png")
plt.show()
```

Graphical Representation

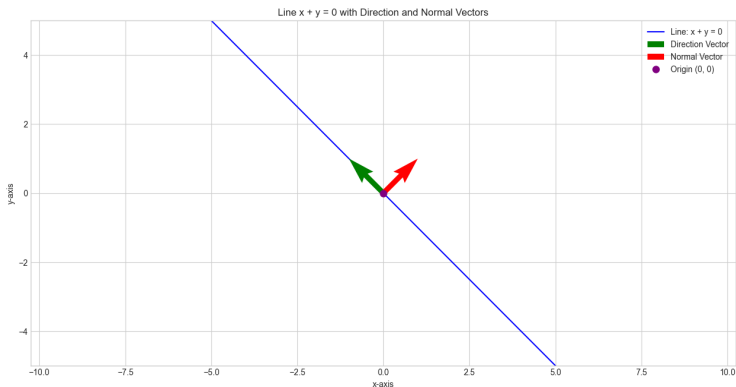


Figure: line $x+y=0$